

FORT BEAUFORT, South Africa

WMO: 687470

Lat: 32.788S

Lon: 26.629E

Elev: 455

StdP: 95.98

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	2.7	3.9	-7.9	2.0	23.3	-5.3	2.5	19.4	11.0	17.0	10.0	22.4	1.8	300	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	35.5	19.8	33.3	19.6	31.4	19.3	22.5	30.2	21.7	29.0	21.1	28.1	4.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.8	25.5	19.6	15.2	24.6	19.0	14.6	23.9	68.7	30.3	65.7	29.0	63.2	28.1	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.5	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	23.3	23.2	21.6	18.7	16.5	13.9	14.3	15.0	16.4	18.3	19.8	21.9
	DBStd	4.79	3.38	3.01	3.28	3.56	3.34	3.59	3.99	3.77	3.48	3.65	3.47	3.34
	HDD10.0	13	0	0	0	0	1	3	6	2	1	0	0	0
	HDD18.3	682	3	2	7	37	77	140	138	120	79	47	24	7
	CDD10.0	3139	413	370	360	262	203	122	138	157	192	257	296	370
	CDD18.3	768	158	139	109	49	21	8	13	16	20	46	69	119
	CDH23.3	5989	1257	963	737	358	198	83	125	204	207	407	536	915
(13)	CDH26.7	2297	559	392	287	119	48	12	17	63	72	146	196	386
(14)	Wind	WSAvg	2.7	2.9	2.7	2.4	2.2	2.3	2.8	2.7	2.7	2.9	3.0	3.0
(15) Precipitation	PrecAvg	519	50	66	72	37	21	17	16	31	30	51	63	63
	PrecMax	846	133	226	287	122	82	84	90	217	105	132	204	203
	PrecMin	250	2	9	6	2	0	0	0	0	2	0	0	1
	PrecStd	149	36	47	60	28	21	21	23	39	28	34	48	52
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	36.8	36.6	33.2	30.6	28.0	28.7	31.9	33.4	35.0	35.5	37.4
		MCWB	20.9	20.6	19.5	18.1	15.6	13.6	12.6	13.5	14.8	17.7	18.5	20.0
	2%	DB	34.7	33.4	32.9	30.2	27.9	25.6	26.5	27.7	29.0	31.0	31.6	33.2
		MCWB	21.1	21.2	19.7	17.7	14.7	12.7	12.3	12.7	15.1	17.5	17.8	19.6
	5%	DB	31.9	31.1	30.0	27.4	25.5	23.6	24.4	25.2	25.5	28.1	28.7	30.7
		MCWB	20.3	21.2	19.4	17.4	14.2	11.8	11.5	12.3	14.9	17.1	17.9	19.7
	10%	DB	29.6	28.8	27.6	24.9	23.1	21.2	22.0	22.2	22.9	25.3	26.3	28.2
		MCWB	20.1	20.5	19.1	16.5	13.8	11.4	11.1	11.9	14.2	16.3	17.4	19.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.2	24.0	22.3	20.5	17.9	15.2	14.2	16.5	18.6	20.3	21.2	22.6
		MCDB	32.4	31.2	29.8	26.7	24.2	21.9	22.6	21.1	25.7	29.6	30.1	31.6
	2%	WB	22.3	22.8	21.3	19.4	16.6	13.6	13.5	15.1	17.2	19.2	19.8	21.4
		MCDB	30.3	29.6	28.6	25.5	23.3	21.6	21.9	21.2	24.3	27.1	28.0	29.9
	5%	WB	21.6	21.9	20.5	18.6	15.7	12.9	12.7	14.2	16.2	18.1	18.9	20.5
		MCDB	29.0	28.2	27.4	24.8	22.0	20.3	20.8	20.3	22.9	24.8	25.7	28.3
	10%	WB	20.9	21.1	19.7	17.6	15.0	12.1	12.0	13.3	15.3	17.1	18.0	19.7
		MCDB	27.9	27.1	26.1	23.3	21.3	19.3	19.4	19.6	21.5	23.3	23.9	26.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	11.1	10.9	11.5	12.2	12.6	12.5	12.4	12.0	11.8	11.9	11.4
		MCDBR	16.3	15.1	15.4	15.0	15.8	15.5	15.1	17.4	17.1	16.9	16.6	16.3
		MCWBR	5.1	4.9	5.1	5.3	5.8	6.2	5.7	6.6	6.7	6.5	6.5	5.8
	5% WB	MCDBR	14.1	13.4	13.0	12.9	12.9	13.0	12.9	12.8	14.3	14.2	14.7	14.6
		MCWBR	5.1	5.2	5.3	5.5	6.0	6.2	6.3	5.9	6.4	6.1	6.4	5.8
(40) Clear-Sky Solar Irradiance	taub		0.369	0.371	0.356	0.328	0.310	0.295	0.298	0.344	0.372	0.383	0.363	0.366
	taud		2.465	2.465	2.482	2.531	2.550	2.560	2.542	2.405	2.325	2.340	2.437	2.467
	Ebn at Noon		969	946	925	899	866	857	871	866	893	924	969	975
	Edh at Noon		118	114	105	89	78	72	77	99	121	128	121	119
(44) All-Sky Solar Radiation	RadAvg		6.85	6.25	5.30	4.16	3.30	2.93	3.16	3.92	5.04	5.92	6.72	7.14
	RadStd		0.36	0.32	0.37	0.22	0.13	0.17	0.18	0.25	0.34	0.42	0.39	0.48

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days		
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.35	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-85	+142

Nomenclature: See separate page